

Open-Source Simulators for Multi-Domain Ocean Robotics: An Annotated Literature Review

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Abstract

This is a working literature review of open-source simulation frameworks usable for ocean robotics in the year 2026. The scope is projects that can represent the marine domain together with adjacent domains (underwater, water surface, air, land), and that satisfy our working definition of a healthy software project: open source, well documented, and supported by a sizable, active community. Projects whose domain-specific layer is itself open source are included even when they sit on closed-source engines (Unreal, Unity, NVIDIA Isaac Sim). Each entry has a short note that says what the project is and why we keep it on the reading list. The aim is a document short enough to keep current and verbose enough that a future contributor can recover the reasoning without reading the underlying papers again.

1 Scope and motivation

We care about simulators that have to serve three use cases. Each one puts different pressure on the underlying tool.

1. **Manual, in-the-loop development and operator training.** The simulator is a visual and dynamic sandbox in which a human iterates on algorithms, missions, or operator procedures. What matters here is rendering, asset availability, and ergonomics. Raw throughput is secondary.
2. **Continuous-integration regression testing for software safety.** The simulator has to run headless, has to be deterministic, has to be fast enough to gate merges, and has to stay stable across OS upgrades. What matters here is reproducibility and packaging. Visual polish does not.
3. **Vectorized training for machine-learning research.** The simulator is treated as a parallelizable environment factory. What matters here is steps per wall-second across many concurrent worlds, GPU acceleration, deterministic resets, and a Python-native interface.

Our domain of interest is *multi-domain ocean robotics*: a single simulated world that may contain assets operating underwater, on the water surface, in the air, and on land, interacting through a shared environment (for example, a USV launching a UAV that overflies a shoreline sensor while an AUV operates below the surface).

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2 Selection criteria

We use three working criteria for inclusion. A project is in scope if it is (i) **open source** with a usable license; (ii) has **documentation good enough that a new user can stand it up without reading the source** (a soft criterion; we note when only the source serves as documentation); and (iii) shows **evidence of a sizable, active community**, taken here as some combination of GitHub stars, recent commit cadence, a maintained forum or chat, and outside citations.

Projects built on proprietary engines (NVIDIA Isaac Sim, Unreal, Unity) are included when the marine layer is itself open source and meets the three criteria above. We **exclude** projects that are one-off academic prototypes without a sustained user base. A small number of borderline cases are noted in Section 5 so the document can be revisited as those communities mature.

3 The landscape

3.1 General-purpose substrate: Gazebo

Modern Gazebo (Koenig and Howard 2004) is the substrate beneath most of the open-source marine stacks we cover. The Open-Robotics-led modern series (Garden, Harmonic, Jetty) replaced the “Ignition” naming after the 2022 trademark issue. Gazebo Classic reached end-of-life in January 2025. The project is a case of large community plus thin documentation: a clean API and a sprawling ecosystem, but learning paths still tend to route through source reading and community channels.

3.2 Gazebo-based marine stacks

The largest cluster of marine simulators builds on the Gazebo family.

UUV Simulator (Manhães et al. 2016) pioneered Gazebo Classic underwater dynamics and is still widely cited, even though the upstream repository was archived in 2023. We keep it here for historical context: it is the genealogical root of most ROS/Gazebo underwater work.

DAVE (Zhang et al. 2022) extends UUV Simulator toward standardized scenes, manipulation, and oceanographic environments. The project is mid-port from Gazebo Classic to modern Gazebo and is led by a group that includes the Naval Postgraduate School, WHOI, and Open Robotics.

VRX (Virtual RobotX) (Bingham et al. 2021) is the most actively maintained marine Gazebo stack we know of. It was originally built for the RobotX Maritime Challenge; modern releases (v3.x, on Gazebo Harmonic and ROS 2 Jazzy) run surface, aerial, and terrestrial assets together in one world. Of the projects in this review, VRX comes closest to delivering the multi-domain story end to end on an open stack.

LRAUV simulation (Player et al. 2023) from MBARI is a production-targeted digital twin used to validate long-duration science missions before deployment. The standalone `osrf/lrauv` fork has been upstreamed into Gazebo and was archived in 2025; ongoing development continues in the MBARI organization.

3.3 Game-engine-based simulators

HoloOcean (Potokar, Ashford, et al. 2022; Potokar, Lay, et al. 2024) is BYU’s Unreal-Engine-based underwater simulator and the most cited non-Gazebo option in marine perception research. Its 2024 *Journal of Oceanic Engineering* release broadens the sensor suite (imaging sonar, DVL, side-scan, multibeam). A 2.0 preview targets Unreal 5.3 with a ROS 2 bridge.

OceanSim (Song et al. 2025) from the Field Robotics Group at the University of Michigan ports the underwater perception problem onto NVIDIA Isaac Sim, inheriting GPU ray tracing and

the Omniverse asset pipeline. It is the most credible candidate in this list for the third use case in Section 1 (vectorized ML training). At the time of writing it is underwater-only, but the underlying engine supports terrestrial and aerial assets, so the gap is one of marine-domain content rather than engine capability.

3.4 Standalone hydrodynamics + rendering

Stonefish (Cieřlak 2019; Grimaldi, Cieřlak, et al. 2025) is a self-contained C++/OpenGL simulator from the University of Girona, with strong manipulator support and ROS 1/ROS 2 wrappers. Its 2025 ICRA paper adds ML-oriented sensors (event camera, thermal, optical flow, visible-light communication) and frames the project explicitly as a substrate for machine-learning research in marine robotics. The license is GPL-3.0, which is more restrictive than the Apache-licensed Gazebo stack and matters for downstream redistribution.

3.5 Hydrodynamics specialist

WEC-Sim (Ruehl et al. 2017) (NREL + Sandia, funded by the US DOE Water Power Technologies Office) is the canonical open-source wave-energy-converter modeling toolbox. It is a MATLAB/Simulink toolchain rather than a robotics simulator, but it is the reference for high-fidelity floating-body hydrodynamics and is worth keeping on the reading list as a verification target for the surface dynamics of robotics simulators.

4 Cross-cutting observations

- **Only VRX currently runs underwater, surface, aerial, and terrestrial dynamics in a single maintained open stack.** Its underwater coverage is shallow next to the perception fidelity of HoloOcean or OceanSim, but the multi-domain integration story is uncommon.
- **Vectorized ML training is best supported on the Isaac-Sim branch (OceanSim) and, to a lesser extent, HoloOcean.** Gazebo’s throughput model does not yet match the thousands-of-parallel-worlds idiom that dominates modern RL.
- **CI/CD adoption favors Gazebo today.** Deterministic headless runs through `gz sim` are mature, and the same binaries are used in development and CI. Unreal- and Unity-based simulators are harder to package for unattended runs.
- **Documentation quality varies a lot.** WEC-Sim, HoloOcean, and VRX have approachable user-facing documentation. Gazebo, DAVE, and Stonefish rely more on source reading and community channels.
- **License diversity matters.** Most of the Gazebo ecosystem and OceanSim are Apache-2.0. HoloOcean is MIT but inherits the Unreal EULA at runtime. Stonefish is GPL-3.0. Downstream redistribution plans need to account for this.

5 Excluded and adjacent

The projects below come up in marine-simulator discussions but do not currently meet the community-size criterion, so they are noted here rather than reviewed.

- **MARUS** (Lončar et al. 2022) — a Unity-based maritime simulator from the University of Zagreb. Promising technical direction but limited adoption outside its originating lab.
- **Project Aquaticus** (Novitzky et al. 2024) — a maritime capture-the-flag testbed originally from MIT and now at USMA West Point, built on MOOS-IvP rather than a 3-D physics simulator. Cite for human-robot teaming context, not as a simulator product.
- **Cosys-AirSim** / **ASVSim** (Jansen et al. 2023; Middelaar et al. 2025) — the University of Antwerp’s active fork of Microsoft AirSim (archived 2022). Mostly aerial, with a 2025 ASV spinoff. Worth monitoring; revisit if the marine layer matures.

A useful umbrella reference that covers many of the same systems is the recent underwater-robotics-simulator review of Aldegheri et al. (2025). We treat it as a companion document rather than a replacement for project-by-project assessment.

6 Contributors and contact

This is a living working review. Contributors are listed below with the area they were responsible for. Please contact the relevant person before making substantive edits in their section so the annotations stay coherent.

Contributor	Contact	Area
Brian Bingham	briansbingham@gmail.com	Overall editor; Gazebo and VRX sections
(open)	(open)	Underwater perception (HoloOcean, OceanSim, Stonefish)
(open)	(open)	Hydrodynamics and surface dynamics (WEC-Sim, LRAUV)
(open)	(open)	ML training infrastructure and vectorized environments

Annotated references

Aldegheri, Stefano et al. (2025). *A Review of Underwater Robotic Simulators*. arXiv: [2504.06245](https://arxiv.org/abs/2504.06245). URL: <https://arxiv.org/abs/2504.06245> (visited on 05/21/2026).

Notes. Recent umbrella survey covering many of the same underwater simulators we discuss here. Treated as a companion reference; the author list and exact affiliations should be verified before citing in a published version of this document.

Bingham, Brian, Carlos Aguero, Michael McCarrin, Joseph Klamo, Joshua Malia, Kevin Allen, Tyler Lum, Marshall Rawson, and Rumman Waqar (2021). “Toward Maritime Robotic Simulation in Gazebo”. In: *Proc. OCEANS 2021 San Diego–Porto*. DOI: [10.23919/OCEANS44145.2021.9705971](https://doi.org/10.23919/OCEANS44145.2021.9705971). URL: <https://github.com/osrf/vrx> (visited on 05/21/2026).

Notes. Reference open-source USV simulator and the backbone of the RobotX Maritime Challenge. Modernization (v3.x on Gazebo Harmonic + ROS 2 Jazzy) is led by Honu Robotics; sponsored by RoboNation. Apache-2.0. The only project we survey that integrates surface, aerial, and terrestrial dynamics in one open, maintained world — which is why it is central to a multi-domain ocean review.

- Cieślak, Patryk (2019). “Stonefish: An Advanced Open-Source Simulation Tool Designed for Marine Robotics, With a ROS Interface”. In: *Proc. OCEANS 2019 Marseille*. DOI: [10.1109/OCEANSE.2019.8867434](https://doi.org/10.1109/OCEANSE.2019.8867434). URL: <https://github.com/patrykcieślak/stonefish> (visited on 05/21/2026).
- Notes.** Self-contained C++/OpenGL simulator from the Underwater Vision and Robotics Lab at the University of Girona (CIRS). Independent of Gazebo, with its own renderer and Bullet physics. GPL-3.0; ROS 1/ROS 2 wrappers via `stonefish_ros`. Included because Stonefish is the most active non-game-engine alternative for high-fidelity underwater simulation.
- Grimaldi, Michele, Patryk Cieślak, et al. (2025). “Stonefish: Supporting Machine Learning Research in Marine Robotics”. In: *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*. arXiv: [2502.11887](https://arxiv.org/abs/2502.11887). URL: <https://github.com/patrykcieślak/stonefish> (visited on 05/21/2026).
- Notes.** Adds ML-oriented sensors (event camera, thermal, optical flow, visible-light communication) and frames Stonefish as an ML research substrate. Cite this rather than the 2019 OCEANS paper when discussing ML use cases.
- Jansen, Wouter, Erik Verreycken, Anthony Schenck, Jean-Edouard Blanqué, Henri Verhulst, and Jan Steckel (2023). *Cosys-AirSim: A Real-Time Simulation Framework Expanded for Complex Industrial Applications*. arXiv: [2303.13381](https://arxiv.org/abs/2303.13381). URL: <https://github.com/Cosys-Lab/Cosys-AirSim> (visited on 05/21/2026).
- Notes.** University of Antwerp’s actively maintained fork of Microsoft AirSim (archived 2022). MIT-licensed; predominantly aerial. Noted because AirSim derivatives are the most common path to marine simulation on Unreal outside of HoloOcean.
- Koenig, Nathan and Andrew Howard (2004). “Design and Use Paradigms for Gazebo, an Open-Source Multi-Robot Simulator”. In: *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*. Sendai, Japan, pp. 2149–2154. DOI: [10.1109/IROS.2004.1389727](https://doi.org/10.1109/IROS.2004.1389727). URL: <https://github.com/gazebo/gz-sim> (visited on 05/21/2026).
- Notes.** Founding paper for Gazebo, now maintained by Open Robotics (post-Ignition rename). Modern Gazebo (Garden, Harmonic, Jetty) is the substrate beneath VRX, DAVE, and LRAUV. Gazebo Classic reached end-of-life Jan 2025. Apache-2.0; very active community via gazebo.org forum, Discord, monthly community meetings. Included as the de-facto open substrate for ROS-based marine simulation.
- Lončar, Igor, Juraj Obradović, Natko Krčmar, Ivan Sitar, Luka Kušnjier, Nikola Mišković, and Zoran Vukić (2022). “MARUS – A Marine Robotics Simulator”. In: *Proc. OCEANS 2022 Hampton Roads*. DOI: [10.1109/OCEANS47191.2022.9977075](https://doi.org/10.1109/OCEANS47191.2022.9977075). URL: <https://github.com/MARUSimulator> (visited on 05/21/2026).
- Notes.** Unity-based maritime simulator from LABUST, University of Zagreb FER. Apache-2.0; gRPC bridge to ROS 1/ROS 2. Technically interesting but with limited adoption outside the originating lab. Noted here so we can revisit if the community grows.
- Manhães, Musa Morena Marcusso, Sebastian A. Scherer, Martin Voss, Luiz Ricardo Douat, and Thomas Rauschenbach (2016). “UUV Simulator: A Gazebo-based package for underwater intervention and multi-robot simulation”. In: *Proc. OCEANS 2016 MTS/IEEE Monterey*. Monterey, CA, USA. DOI: [10.1109/OCEANS.2016.7761080](https://doi.org/10.1109/OCEANS.2016.7761080). URL: https://github.com/uuvsimulator/uuv_simulator (visited on 05/21/2026).
- Notes.** The genealogical root of the open-source Gazebo+ROS underwater stack. Repository archived Feb 2023 and the project is unmaintained, but it is the upstream of DAVE and is still the most commonly cited paper for AUV simulation in Gazebo. Apache-2.0. Kept for historical and didactic value, not as a recommended current tool.
- Middelaar, Robin van et al. (2025). *ASVSim: A Simulation Framework for Autonomous Surface*

Vehicles Built on Cosys-AirSim. arXiv: 2506.22174. URL: <https://arxiv.org/abs/2506.22174> (visited on 05/21/2026).

Notes. *Surface-vehicle spinoff of Cosys-AirSim. Too new (2025) to claim community traction; included so we can revisit the Unreal+ASV story in a future pass.*

Novitzky, Michael, Charles Pippin, Thomas R. Collins, Paul Fitzpatrick, Alexander Holness, and Henrik Schmidt (2024). *Evaluating Collaborative Autonomy in Opposed Environments Using Maritime Capture-the-Flag Competitions*. arXiv: 2404.17038. URL: <https://oceanai.mit.edu/aquaticus/> (visited on 05/21/2026).

Notes. *Project Aquaticus is a maritime capture-the-flag testbed originally developed at MIT and migrated to USMA West Point in 2019. Built on MOOS-IvP rather than a 3-D physics simulator; cite for human-robot teaming context, not as a comparable simulator product.*

Player, Timothy R., Arjo Chakravarty, Mabel M. Zhang, Ben Yair Raanan, Brian Kieft, Yanwu Zhang, and Brett Hobson (2023). “From Concept to Field Tests: Accelerated Development of Multi-AUV Missions Using a High-Fidelity Faster-than-Real-Time Simulator”. In: *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*. London, UK. URL: https://github.com/mbari-org/gz_lrauv (visited on 05/21/2026).

Notes. *Production digital twin of MBARI’s Tethys-class long-range AUV, used to validate multi-day science missions before deployment. Demonstrates Gazebo (modern) for long-duration AUV ops. The standalone `osrf/lrauv` fork was archived in Mar 2025 after upstreaming; ongoing work continues in the MBARI org. Apache-2.0. Included as an exemplar of production use of an open simulator.*

Potokar, Easton, Spencer Ashford, Michael Kaess, and Joshua G. Mangelson (2022). “HoloOcean: An Underwater Robotics Simulator”. In: *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*. Philadelphia, PA, USA. DOI: 10.1109/ICRA46639.2022.9812388. URL: <https://byu-holoocean.github.io/holoocean-docs/> (visited on 05/21/2026).

Notes. *Initial release of BYU FRoStLab’s Unreal-Engine underwater simulator, built on top of Microsoft Holodeck. Python-first API. MIT license on the client side (Unreal EULA applies to the engine). Kept because HoloOcean is the most widely cited non-Gazebo underwater simulator in marine perception research.*

Potokar, Easton, Kalin Lay, Kalebh Norman, Devin Benham, Spencer Ashford, Robert Peirson, and Joshua G. Mangelson (2024). “HoloOcean: A Full-Featured Marine Robotics Simulator for Perception and Autonomy”. In: *IEEE Journal of Oceanic Engineering*. DOI: 10.1109/JOE.2024.3403060. URL: <https://froslab.byu.edu/holoocean-underwater-simulator> (visited on 05/21/2026).

Notes. *Expanded sensor suite (imaging sonar, side-scan, multibeam, DVL) and improved hydrodynamics. The reference paper to cite for current HoloOcean capabilities. A 2.0 preview targeting Unreal 5.3 and ROS 2 appeared in late 2025 (arXiv:2510.06160) but is not yet a stable release.*

Ruehl, Kelley, Carlos Michelen, Bret Bosma, and Yi-Hsiang Yu (2017). “WEC-Sim: An Open-Source Tool for Simulating Wave Energy Converters”. In: *Proc. 12th European Wave and Tidal Energy Conf. (EWTEC)*. Cork, Ireland. URL: <https://wec-sim.github.io/WEC-Sim/> (visited on 05/21/2026).

Notes. *NREL + Sandia, funded by US DOE Water Power Technologies Office. MATLAB/Simulink/Simscape Multibody toolchain for floating-body hydrodynamics; not a robotics simulator. Apache-2.0; v7.0 released Aug 2025; runs an annual training course and active forum. Included as the domain standard for surface hydrodynamics — useful as a verification reference, not as a primary robotics simulator.*

Song, Jingyu, Haoyu Ma, Onur Bagoren, Advait V. Sethuraman, Yiting Zhang, and Katherine A. Skinner (2025). “OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework”. In: *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*. arXiv: 2503.01074. URL: <https://github.com/umfieldrobotics/OceanSim> (visited on 05/21/2026).

Notes. *Brings underwater simulation onto NVIDIA Isaac Sim / Omniverse, with GPU ray tracing*

and a USD asset pipeline. Lead group: Field Robotics Group (FRoG), University of Michigan; PI Katherine A. Skinner. Apache-2.0; very active (v4.5 released Sept 2025, Isaac Sim 5.0 compatible). The strongest current candidate for vectorized ML training in marine perception. Underwater-only today, but the underlying engine extends naturally to surface, aerial, and terrestrial assets.

Zhang, Mabel M., Woen-Sug Choi, Jonathan Herman, Duane Davis, Carson Vogt, Michael McCarrin, Vaibhav Vijay, Dharini Dutia, William Lew, Steven Peters, and Brian Bingham (2022). “DAVE Aquatic Virtual Environment: Toward a General Underwater Robotics Simulator”. In: *Proc. IEEE/OES Autonomous Underwater Vehicles Symposium (AUV)*. Singapore. arXiv: [2209.02862](https://arxiv.org/abs/2209.02862). URL: <https://github.com/Field-Robotics-Lab/dave> (visited on 05/21/2026).

Notes. *Extends UUV Simulator with standardized environments (bathymetry, currents), manipulation, and oceanographic data. Multi-institutional (NPS, WHOI, Open Robotics) and a port to modern Gazebo is in progress. Apache-2.0; documentation site at <https://field-robotics-lab.github.io/dave.doc/>. Included as the most ambitious extension of the UUV Sim lineage.*